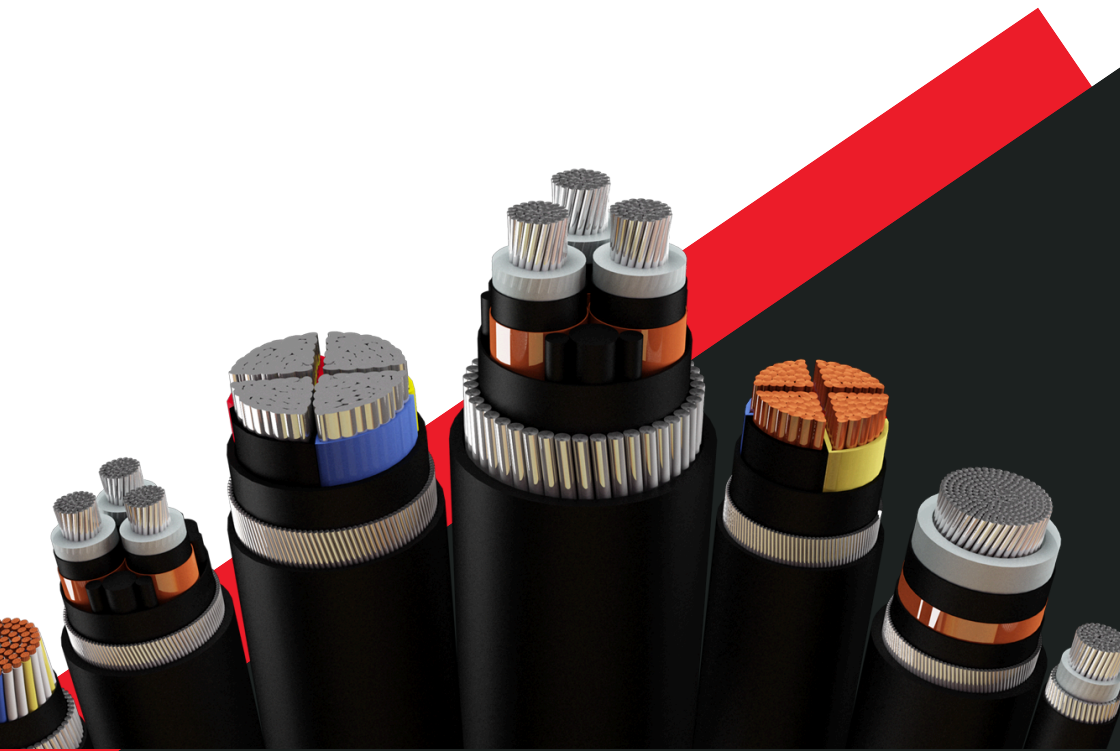




Eight Decades of Trust & Innovation

DEFENSE AND AEROSPACE CABLES







ABOUT US

- Nicco Cables is a 80 year old brand based in India
- Nicco specializes in manufacturing wide range of Wires and Cables
- Nicco has an in-house R&D facility
- Nicco specializes in Compound manufacturing and has a dedicated team for Compounds
- Nicco is the first Company in India to install a 3 MeV Electron Beam Plant form USA for manufacturing irradiated cables
- The manufacturing plant covers an area of 450,000 sq.ft.
- National Accreditation Board for Testing and Calibration Laboratories (NABL) accredited Testing laboratory at plant

NICCO PRODUCES A WIDE RANGE OF CABLES

- LT Power & Control Cables
- HT Power & Control Cables (Upto 33 kv)
- Solar and Windmill cables
- Elastomeric & Silicon Cables (Upto 15 kv) for various application
- Flexible Trailing Cables - H07-RNF
- Ethernet Polyurethane Cables
- Rolling Stock Cables
- Fire Survival Cables (Including Nuclear Reactor)
- TREE WIRE / SPACER Cables - 3 layer Track Resistant (upto 35kv)
- Overhead transmission Conductors (AAAC, ACSR, AAC, ACAR , AL-59)
- Automotive Cables
- Cable Harnessing

SHIPBUILDING CABLES

General Cables:

Marine Shipboard & Offshore cables are low smoke, halogen-free, flame-retardant, and fire-resistant cables that are commonly used for lighting, control circuits, communication applications, shipboard wiring, offshore platforms for electric power distribution to electric panels, and other at-sea installations for oil drilling and production. The increased need for low weight and increased efficiency is driving up requirements for maritime cables in the shipbuilding industry.

Applications:

There are several uses for marine cable in the maritime environment. It is applicable to feeder vessels, gas tankers, reefers, cruise ships, drill ships, bulk carriers, tankers, and other types of ships.

Marine cables shall possess following properties:

- High insulation resistance
- High resistance to abrasion, impact & compression
- Resistant to minerals, grease & chemicals
- Oil and flame-resistant
- Resistance to humidity and water absorption
- Resistance to ozone & UV radiation
- Fire Survival property
- Low halogen, low smoke & low toxic Cables
- Flame Retardant (Category – A, B & C)
- Pressure Tight Cables up to 72 Bar
- EMI/EMC compatibility

*Other specific properties as per customer requirements

DIFFERENT TYPES OF SHIPBUILDING/MARINE CABLE

- Power Cable
- Pressure Tight Cable (PT-10, PT-45, PT-60, PT-72)
- Degaussing Cable
- EMC & VFD Cable
- Silicon Cable (Extruded & T-Type)
- Instrumentation Cable
- Shore Supply Cable
- Ethernet/Data Cable (CAT-5e, CAT-6, CAT-6A, CAT-7)
- FMSGSSGO, LFMGSSGO, FMSGSSGO
- Radio Frequency (RF) Cable
- Russian Cable
- TOW Cable
- PTFE, ETFE & FEP Cables

SHIPPING CABLES STANDARDS

- IEC 60092-350
- IEC 60092-353
- IEC 60092-376
- BS 6883 & BS 7917
- VG 95218 (SERIES)
- IEC 60092 (SERIES)
- GOST- 7866, 11326
- NES / DEF STAN - 526 & 527
- EED-50-12 (EPR INSULATED, EVA SHEATHED)
- EED-50-13 (SILICON INSULATED EVA SHEATHED)
- EED-57-03 (PRESSURE TIGHT, PT-03, PT-10)
- EED-57-04 (PRESSURE TIGHT, PT-04, PT-45)
- DMDE STANDARD

AEROSPACE, DEFENSE & COMMUNICATION CABLES

MAJOR SEGMENTS

- **Defense Application**
 - Navy vessels, Defense Networking and communication equipment, tanks assemblies and ATV , Defense Automotive
- **Aerospace Application**
 - ISRO Middle module wiring (all GLSV, PLSV, Rockets), HAL - Air frame wire, DRDO & DRDL - All aerospace application products.
- **Automotive Industries**
 - Class-D, E, F,G, H ISO 6722 wires and cables, EV Motor Assemblies being used by OLA and other EV two & four wheeler manufacturers, Oxygen sensor, all engine frame wiring etc.
- **Government/PSUs/Private Customer**
 - All high temperature applications, Transformers, Furnace, Motors along with High frequency application - Phase shifter, radar, satellite etc. (Generally being used by BEL for all microwave electronics products)

PRODUCTS UNDER DEVELOPMENT

- **Corona Resistant High Voltage Wires and Cables**
 - 4 KV to 22 KV AC, 9 KV to 50 KV DC, AWG 28 to AWG 10, Corona resistant PTFE, Cryogenic rated to +260°C
- **Equipment/Hook-up Wires**
 - PTFE (Tape Wrapped & Sintered), PTFE (Extruded), ETFE (Light & Medium Weight), FEP, PFA, -200°C to +260°C, Silver Plated Copper, Nickel Plated Copper, Tin Plated Copper

- **Multicore/ Hybrid Wires & Cables**

- Multi-core cables ranging from shielded twisted pairs of AWG 34 to AWG 12. Multiple number of pairs, triads, quads, coaxial cables etc. combined into one large cable, with or without shielding on individual pairs/triads/quads, as well as with overall shielding. Jackets can be PTFE, FEP, ETFE or Varnished Fibre Glass (VFG), for high temperature requirements

- **RF Co-axial Cables**

- RG 141, 142, 144, 165, 178, 179, 180, 187, 188, 195, 196, 225, 302, 303, 304, 316, 316DS, 400, 403 and RG 404. They meet or exceed US MIL-DTL-17 or MIL-C-17

- **Heating Cables**

- Underfloor heating cables are an ideal solution for installing underfloor heating in both large or small projects and are a reliable and effective underfloor heating solution that works with tiles and many other floor coverings.

- **Thermocouple Wires & Extension**

- PTFE/FEP Insulated thermocouple combinations offered with 7, 19 strands of 0.2 mm or 0.25 mm dia strands. $\pm 0.75\%$ temperature tolerance as per IEC 584-2

No.	Product / Service	Military Specification	Detailed Specification Sheet
1	Polyimide (Kapton®) Insulated Wire	MIL-DTL-81381	/7, /8, /9, /10, /11, /12, /13, /14, /17, /18, /19, /20, /21, /22
2	FP / Polyimide / FP (Hybrid) Tape & FP Tape Insulated (Composite) Wire [FP=Modified PTFE]	MIL-DTL-22759	/80A, /81A, /82A, /83A, /84A, /85A, /86A, /87A, /88A, /89A, /90A, /91A, /92A.
3	Fire Resistant, Flight Critical Wires	MIL-DTL-25038	/1, /3, & Equivalent to Filotex 1100, 1200, 3000, 3100
4	Cable of above "Wires" • Single Core / Multicore • Twisted / Without Twisted • Shielded / Without Shielded • Jacketed / Without Jacketed	MIL-DTL-27500	MR, MS, MT, MV, MW, MY, NA, NB, NE, NF, NG, NH, NK, NL, WB, WC, WE, WF, WG, WH, WJ, WK, WL, WM, WN, WP, WR, JA, JF.
		Jacket Style	11/61, 12/62, 06/56, 07/57, 24/74
6	"K" Type Thermocouple Extension Wire & Cables	MIL-W-5846C, BS-4937, MIL-DTL-27500	All items of "K" Type
7	Aromatic Polyimide Lacquered Wire ("Magnet Wire")	NEMA : MW-1000, IEC 317-7	/16C Part 7
8	Russian Type Cables: BIN, BPDO, BPDOU, PTL-200 / 250, BIF, BIFM, BFS, BPVL, BPVLM, MGTFL, MGSHV, MS 16/26/36-P / MS-16/26/36-13-33	Applicable Russian Specifications	All items
9	DOLN Type Cable (For Dornier)	Applicable Dornier Specifications	All items

Product Construction

These wires and cables are made from

- **Conductor**
 - Solid or Multi - strand Silver / Nickle / Tin plated Copper / Copper Alloy / Copper Covered Steel Wire
- **Insulation**
 - PTEE, Polyimide, Composite of PTFE & Polyimide, Siliceous Fiber glass Silicon etc.
- **Shield**
 - Braids of Silver / Nickle / Tin plated Copper wire
- **Jacket**
 - Polyimide, PTFE, PTFE + Polyimide, Fiberglass, Normex etc.

The SPC / NPC / TPC conductor is manufactured in our group company SPC Products to ensure complete quality control over final product. Our product range meets the design and technical requirements of below mentioned specifications.

Major Approvals

- **CEMILAC**
 - **RCMA**
 - **DGCA**
 - **ISRO**
- (Centre of Military Airworthiness & Certification)
 - (Regional Centre for Military Airworthiness) for on-board Military Aircrafts / Helicopters use
 - (Director General of Civil Aviation) with Release Note Signatory for on-board Civil Aircrafts / Helicopters Use
 - (Indian Space Research Organization) for on-board Satellite & Launch vehicle Use

Major Certification & Covered Specification

- **Quality Management System**
 - **Product Certification**
 - **Major product specification covered**
- ISO 9001, AS 9100D (If interested in Air-borne Market), DGAQA, LCSO (Capacity Assessment), ISRO & HAL Qualification.
 - LCSO, CEMILAC, C-DOT, Q-DAS (ISRO product qualification), MIL-22759 (if required)
 - JSS 51034, JSS 51038, MIL/DTL-16878, MIL/DTL-27500 H, SAE 22759 / Various detailed specifications, UL AWM styles, Q-DAS, ANSP (Marine) , IEC 60096, JSS 6124 Series.

EXISTING CERTIFICATIONS

ISO 9001 / 14001 / 45001

Directorate of Quality Assurance (Navy) [DQAN] Registration Certificate

Defense Research and Development Laboratory (DRDL) Registration Certificate

Integrated Headquarters of Ministry of Defense (IHQ/DEE/MOD)

Research Designs and Standards Organization (RDSO)

American Bureau of Shipping (ABS)

Indian Register of Shipping (IRS)

Underwriters Laboratories (UL)

Central Power Research Institute (CPRI)

Bureau of Indian Standards (BIS)

International Railway Industry Standard (IRIS)

National Accreditation Board for Testing and Calibration Laboratories (NABL)

Det Norske Veritas (DNV)

CERTIFICATES UNDER PROGRESS

Bureau Veritas (BV)

Lloyd's Register Group Limited (LR)





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